Sanjay Krishna T N

Cloud & DevOps Engineer | AWS & Azure

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SUMMARY

Cloud & DevOps Engineer with hands-on experience automating AWS & Azure infrastructure using Terraform and building CI/CD pipelines with GitHub Actions. Experienced in deploying and managing containerized and multi-stack applications on Linux servers using Docker and Nginx. Served as the sole DevOps engineer across multiple projects, managing cloud infrastructure, deployment automation, Linux-based application hosting, monitoring, and operational reliability.

SKILL SETS

Programming Language: Python, Bash/Shell Scripting.
Cloud Platforms: AWS (EC2, S3, IAM, Lambda, CloudWatch), Azure
DevOps & Automation: Terraform, GitHub Actions, Docker, CI/CD Pipelines, SonarQube.
Linux & Operations: Linux Administration, Systemd, Log Analysis.
Monitoring & Reliability: Prometheus, Grafana, CloudWatch, Health Checks, Automated Failover.

WORK EXPERIENCE

Cloud & DevOps Engineer – TSK Automations (On-Site) *MAR 2026 - PRESENT*

- Served as the sole DevOps engineer across 4+ projects, managing cloud infrastructure, CI/CD automation, and deployment operations.
- Implemented containerized application deployments using Docker and managed Linux-based hosting environments using Nginx and systemd.
- Set up Prometheus and Grafana dashboards across 4+ services, replacing manual log checks with real-time alerting and service health visibility.
- Built automated health-check workflows that detect service failures and trigger failover without manual intervention, removing on-call toil from routine outages.
- Diagnosed and resolved production infrastructure issues — port conflicts, firewall misconfigurations, and DNS failures — restoring application availability during live incidents.
- Integrated SonarQube into CI/CD workflows to improve code quality validation and deployment readiness.
- Standardized Docker image build, tagging, and promotion workflows across dev and production, eliminating environment drift between deployments.

Cloud & DevOps Engineer – Staunch Technologies Pvt. Ltd. (On-Site) *SEP 2025 - FEB 2026*

- Served as the sole DevOps contributor for a 6-month engagement, managing cloud infrastructure, CI/CD automation, and deployment operations across 4+ projects.
- Built and maintained GitHub Actions pipelines, automating application build and deployment workflows for development and production environments.
- Provisioned and managed AWS and Azure infrastructure using Terraform, ensuring consistent and reproducible environments across development and production workloads.
- Configured Nginx reverse proxies and systemd services for application hosting while supporting end-to-end deployment operations.
- Diagnosed and resolved HTTP 502 errors and environment misconfigurations during live releases, unblocking deployments without rolling back.
- Owned full release cycle from environment provisioning to production deployment, ensuring consistent application availability across all project handoffs.

PROJECTS

AUTOMATED CLOUD FAILOVER & RECOVERY SYSTEM *APRIL 2026 - MAY 2026*

- Designed and implemented an automated failover solution using AWS Lambda, health checks, and DNS-based traffic switching.
- Developed monitoring workflows to continuously validate application availability and trigger failover actions during service outages.
- Automated recovery validation and failback mechanisms to reduce manual intervention and improve service reliability.
- Implemented DNS automation and Lambda-triggered health checks to enable seamless primary-to-secondary environment transitions.

INTENT BASED CLOUD RESOURCE PROVISIONING *NOV 2025 - DEC 2025*

- Developed a self-service cloud management platform using Streamlit and FastAPI that enables users to provision AWS resources without direct console access.
- Automated EC2 and S3 provisioning through Terraform execution workflows with role-based approval and OTP authentication mechanisms.
- Integrated user/admin management, request validation, infrastructure automation, and SSH-based resource management into a unified dashboard.
- Implemented export and approval workflows to simplify cloud resource provisioning for non-technical users.
- Reduced resource provisioning from manual cloud-console operations to a guided request-and-approval workflow.

OIL SPILL DETECTION & MONITORING SYSTEM *JAN 2026 - MAR 2026*

- *Academic research project — demonstrates ML engineering and model deployment capability.*
- Developed an end-to-end oil spill detection system using YOLO, CNN, and U-Net models for detection, classification, and segmentation of oil spill regions from satellite imagery.
- Built a Streamlit-based application for image analysis, prediction visualization, and real-time interaction with the trained deep learning models.
- Implemented dataset preprocessing, model training, evaluation workflows, and segmentation overlays to improve spill identification and monitoring capabilities.
- Designed the solution with a lightweight multi-stage architecture focused on balancing detection accuracy, computational efficiency, and deployment readiness.

EDUCATION

Prathyusha Engineering College, Thiruvallur, Tamil Nadu
B.TECH - Information Technology Nov 2022 - April 2026
CGPA : 8.43 / 10 (as of 8th Semester)

CERTIFICATIONS

- **TCS** - iON National Qualifier Test
- **NPTEL -** Cloud Computing